

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.

Large language models store factual knowledge in their parameters, but this knowledge is static and can become outdated. Retrieval-augmented generation combines a pretrained language model with a non-parametric memory accessed through a retriever, allowing the model to look up relevant documents at inference time rather than relying solely on memorized weights. This approach improves factual accuracy and allows knowledge to be updated without retraining the underlying model.

FAISS, developed by Facebook AI Research, is a library for efficient similarity search and clustering of dense vectors. It supports both exact search via flat indexes and approximate nearest neighbor search via inverted file indexes and hierarchical navigable small world graphs. FAISS is widely used as the vector store component in retrieval-augmented generation pipelines because it scales to billions of vectors while remaining fast on both CPU and GPU hardware.